MICHAEL A. DALLARA

Software Engineer — AI Applications & Real-Time Simulation

C# / .NET · Unity · Anthropic Claude API · agentic & structured-output pipelines · physics simulation

San Mateo, CA · 650-302-3109 · mikedallara2@gmail.com

linkedin.com/in/mike-d-89a75274 · github.com/Biscuittlol/DallaraMikePortfolio · youtube.com/@MikeyDGames101

SUMMARY

Software engineer with 8+ years building interactive 3D applications and real-time simulations in C#, now focused on LLM-driven systems. Designed and built an agentic pipeline on the Anthropic Claude API that generates structured, reproducible, fully editable 3D models from natural language at runtime, and shipped two commercial products end-to-end. Self-taught and product-driven — strongest where AI meets real, interactive systems.

KEY ACHIEVEMENTS

- Built an LLM→deterministic-engine pipeline (Anthropic Claude API) that turns plain language into reproducible, fully editable 3D models — 30+ labeled, code-addressable parts per object, from a single bolt to an aircraft carrier.
- Shipped two commercial titles independently on Steam, plus 5+ substantial original systems spanning AI/agent behavior, real-time physics, procedural generation, and custom tooling.
- 8+ years in C#/Unity designing data-driven, event-driven architectures with unit-tested core logic — no formal CS degree; the work is the credential.

TECHNICAL SKILLS

AI / LLM: Anthropic Claude API, agentic multi-step pipelines, structured/constrained generation (JSON-schema outputs), prompt engineering, adaptive extended thinking, deterministic & reproducible generation, schema validation

Languages: C#, .NET (8+ yrs); Python (in progress)

Simulation & 3D: real-time physics modeling, computational geometry, procedural generation, agent behavior (finite-state machines, steering, schooling), NavMesh

Architecture: data-driven (ScriptableObject) design, event-driven decoupling, manager/service singletons, object pooling & streaming, separation of concerns, unit testing (Unity Test Framework)

Engines & Tools: Unity 6 (URP), ProBuilder, Shader Graph, custom EditorWindow tooling, new Input System, Newtonsoft JSON

Platforms: Windows, Android; Oculus, SteamVR, OpenXR

Tooling: Git, PlasticSCM, Visual Studio, Blender

SELECTED PROJECTS

AI Runtime 3D Generator — 2026 – ongoing

LLM application · C#, Unity 6, Anthropic Claude API (Opus)

- Solved the reproducibility problem in generative 3D: constrained an LLM to a typed JSON schema executed by a hand-written deterministic C# engine, so the model never touches geometry and one prompt always rebuilds the identical, verifiable result.
- Designed a two-pass agentic pipeline (build structure → assign materials) on Claude Opus with adaptive extended thinking, plus a calibrated prompt-to-engine contract that holds across an open-ended domain — a single bolt to an aircraft carrier.
- Made output a programmable object graph, not a static mesh: each generation decomposes into 30+ named, individually addressable parts that can be edited, animated, or driven by code.
- Hardened for production: input/schema validation (rejects degenerate sizes, detects circular references), resilient JSON extraction, and EditMode unit tests on the core geometry math.

Real-Time Fishing & Crabbing Simulation — ongoing

Physics & agent simulation · C#, Unity 6 (URP); ~110 scripts, ~21k LOC

- Engineered a real-time multi-body physics simulation in which every actor resolves in one shared force unit, producing emergent, physically honest behavior tuned by parameters rather than scripted constants.

- Modeled agent stamina as a function of resistance rather than time — yielding emergent, skill-based dynamics — atop a constraint-based deformable-rod solver (integrated curvature arc, Verlet pendulum).
- Architected a data-driven ecology simulation: declarative agent definitions (habitat, depth, schooling, season, time-of-day) reconciled each cycle by a spawner that places agent groups via terrain raycasts, with pooled/streamed agents and cached sampling for performance.
- Wrote a custom procedural terrain-generation editor tool (domain-warped noise, parameterized authoring, automatic slope/depth texturing, non-destructive regeneration).

Shipped Commercial Titles — Safari Hunter & A Terrible Place (Steam)

End-to-end delivery · C#, Unity, Oculus/SteamVR

- Took two VR titles from empty project to commercial Steam release independently — Safari Hunter (~650 integrated scripts, ~150 original) and A Terrible Place (a 9-mission narrative campaign plus a survival mode).
- Designed an extensible agent-AI framework — one abstract base with prey/predator specialization scaling to 29 species via an 11-state machine on NavMesh — plus rarity-weighted, resource-aware spawning that maintains a self-replenishing population.
- Delivered the supporting stack end-to-end: binary save/load, async scene streaming, event-driven inter-system architecture, and a reusable noise-based dissolve shader.

Earlier projects

- LSAT Prep Simulator — a complete standalone desktop app (Unity, C#): data-driven test engine from a typed question bank, single-controller multi-screen UI, and on-disk persistence for cross-session history and best scores.
- RTS / RPG prototypes (Wildkins, Battle Knights) — agent behavior via state machines, modular inventory/economy systems, unit formations, and extended pathfinding for large-scale control.
- Glass Measurement App (Android) — a commercial calculation tool with exportable reports, used by construction teams on projects including Salesforce Tower.

EXPERIENCE

Mikey D Games Mar 2017 – Present

Independent Software Developer (C# / Unity) · Remote

Independent design, architecture, and engineering of the AI tooling, simulations, and shipped titles above — full lifecycle across systems code, AI/agent behavior, custom tooling, UI, and deployment (Steam, Oculus, Android).

Mission Glass Oct 2011 – Present

Union Glazier · San Mateo, CA

Commercial glazing on large-scale construction; independently built and shipped an Android measurement tool (above) to streamline on-site calculations.

Genomic Health Jan 2008 – Dec 2009

Robotic Process Automation Intern · San Mateo, CA

Automated laboratory-robot workflows (Tecan / EVOware) with QA and troubleshooting to keep operations precise and consistent.

CONTINUOUS LEARNING

- Complete C# Unity Developer Course — Udemy (2017)
- Advanced C# Scripting for Unity — Udemy (2017)
- Ongoing self-directed study in LLM application development and AI systems.